However, the Riemannian exponential mapping $\exp_p(\cdot)$ is often computationally intractable since it requires to solve a system of second-order differential equations [38, 1]. Thus instead of using $\exp_p$, we shall rather use a computable *Euclidean retraction* $R : T_p \to \mathbb{R}^D$ of the exponential map expressed in a local θ-coordinate system as:

$$\text{RetG}: \quad \theta_{t+1} = R_{\theta_t}\left(-\alpha_t \nabla_\theta L_\theta(\theta_t)\right). \tag{175}$$

Using the retraction [1] $R_p(v) = p + v$ which corresponds to a first-order Taylor approximation of the exponential map, we recover the *natural gradient descent* [6]:

$$\text{NG}: \theta_{t+1} = \theta_t - \alpha_t g_\theta^{-1}(\theta_t)\nabla_\theta L_\theta(\theta_t). \tag{176}$$

The *natural gradient* [6] (NG)

$${}^{\text{NG}}\nabla L_\theta(\theta) := g_\theta^{-1}(\theta)\nabla_\theta L_\theta(\theta) \tag{177}$$

encodes the *Riemannian steepest descent* vector, and the natural gradient descent method yields the following update rule

$$\text{NG}: \theta_{t+1} = \theta_t - \alpha_t \, {}^{\text{NG}}\nabla L_\theta(\theta_t). \tag{178}$$

Notice that the natural gradient is a *contravariant vector* while the ordinary gradient is a *covariant vector*. Recall that a covariant vector $[v_i]$ is transformed into a contravariant vector $[v^i]$ by $v^i = \sum_j g^{ij} v_i$, that is by using the dual Riemannian metric $g_\eta^*(\eta) = g_\theta(\theta)^{-1}$. The natural gradient is *invariant* under an invertible smooth change of parameterization. However, the natural gradient *descent* does *not* guarantee that the locations θ_t's always stay on the manifold: Indeed, it may happen that for some t, $\theta_t \notin \Theta$ when $\Theta \neq \mathbb{R}^D$.

Property 5 ([21])**.** *The natural gradient descent approximates the intrinsic Riemannian gradient descent using a contravariant gradient vector induced by the Riemannian metric tensor g. The natural gradient is invariant to coordinate transformations.*

Next, we shall explain how the natural gradient descent is related to the *mirror descent* and the *ordinary gradient* when the Riemannian space Θ is dually flat.

4.1.3 Natural gradient in dually flat spaces: Connections to Bregman mirror descent and ordinary gradient

Recall that a dually flat space (M, g, ∇, ∇^*) is a manifold M equipped with a pair (∇, ∇^*) of dual torsion-free flat connections which are coupled to the Riemannian metric tensor g [8, 77] in the sense that $\frac{\nabla + \nabla^*}{2} = {}^{LC}\nabla$, where ${}^{LC}\nabla$ denotes the unique metric torsion-free Levi-Civita connection.

On a dually flat space, there exists a pair of dual global *Hessian structures* [120] with dual canonical Bregman divergences [23, 8]. The dual Riemannian metrics can be expressed as the Hessians of dual convex potential functions F and F^*. Examples of Hessian manifolds are the *manifolds of exponential families* or the *manifolds of mixture families* [86]. On a dually flat space induced by a strictly convex and C^3 function F (Bregman generator), we have two dual global coordinate system: $\theta(\eta) = \nabla F^*(\eta)$ and $\eta(\theta) = \nabla F(\theta)$, where F^* denotes the Legendre-Fenchel convex conjugate function [70, 71]. The Hessian metric expressed in the primal θ-coordinate system is $g_\theta(\theta) = \nabla^2 F(\theta)$, and the dual Hessian metric expressed in the dual coordinate system is $g_\eta^*(\eta) = \nabla^2 F^*(\eta)$. Crouzeix's identity [32] shows that $g_\theta(\theta) g_\eta(\eta) = I$, where I denotes the $D \times D$ matrix identity.

The ordinary gradient descent method can be extended using a *proximity function* $\Phi(\cdot, \cdot)$ as follows:

$$\text{PGD}: \quad \theta_{t+1} = \arg\min_{\theta \in \Theta} \left\{ \langle \theta, \nabla L_\theta(\theta_t) \rangle + \frac{1}{\alpha_t} \Phi(\theta, \theta_t) \right\}. \tag{179}$$

When $\Phi(\theta, \theta_t) = \frac{1}{2}\|\theta - \theta_t\|^2$, the PGD update rule becomes the ordinary GD update rule.

Consider a Bregman divergence [23] B_F for the proximity function Φ: $\Phi(p,q) = B_F(p:q)$. Then the PGD yields the following *mirror descent* (MD):

$$\text{MD}: \quad \theta_{t+1} = \arg\min_{\theta\in\Theta} \left\{ \langle \theta, \nabla L(\theta_t)\rangle + \frac{1}{\alpha_t} B_F(\theta:\theta_t) \right\}. \tag{180}$$

This mirror descent can be interpreted as a natural gradient descent as follows:

Property 6 ([112])**.** *Bregman mirror descent on the Hessian manifold $(M, g = \nabla^2 F(\theta))$ is equivalent to natural gradient descent on the dual Hessian manifold $(M, g^* = \nabla^2 F(\eta))$, where F is a Bregman generator, $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$.*

Indeed, the mirror descent rule yields the following natural gradient update rule:

$$\begin{aligned} \text{NG}^*: \eta_{t+1} &= \eta_t - \alpha_t (g^*_\eta)^{-1}(\eta_t) \nabla_\eta L_\theta(\theta(\eta_t)), & (181)\\ &= \eta_t - \alpha_t (g^*_\eta)^{-1}(\eta_t) \nabla_\eta L_\eta(\eta_t), & (182) \end{aligned}$$

where $g^*_\eta(\eta) = \nabla^2 F^*(\eta) = (\nabla^2_\theta F(\theta))^{-1}$ and $\theta(\eta) = \nabla F^*(\theta)$.

The method is called mirror descent [24] because it performs that gradient step in the *dual space* (ie., mirror space) $H = \{\eta = \nabla F(\theta) \ : \ \theta \in \Theta\}$, and thus solves the inconsistency contravariant/covariant type problem of subtracting a covariant vector from a contravariant vector of the ordinary GD (Eq. 169).

Let us prove now the following property of the natural gradient in a dually flat space or Bregman manifold [77]:

Property 7 ([137])**.** *In a dually flat space induced by potential convex function F, the natural gradient amounts to the ordinary gradient on the dually parameterized function: ${}^{\text{NG}}\nabla L_\theta(\theta) = \nabla_\eta L_\eta(\eta)$ where $\eta = \nabla_\theta F(\theta)$ and $L_\eta(\eta) = L_\theta(\theta(\eta))$.*

Proof. Let (M, g, ∇, ∇^*) be a dually flat space. We have $g_\theta(\theta) = \nabla^2 F(\theta) = \nabla_\theta \nabla_\theta F(\theta) = \nabla_\theta \eta$ since $\eta = \nabla_\theta F(\theta)$. The function to minimize can be written either as $L_\theta(\theta) = L_\theta(\theta(\eta))$ or as $L_\eta(\eta) = L_\eta(\eta(\theta))$. Recall the chain rule in the calculus of differentiation:

$$\nabla_\theta L_\theta(\theta) = \nabla_\theta(L_\eta(\eta(\theta))) = (\nabla_\theta \eta)(\nabla_\eta L_\eta(\eta)). \tag{183}$$

Thus we have:

$$\begin{aligned} {}^{\text{NG}}\nabla L_\theta(\theta) &:= g_\theta^{-1}(\theta) \nabla_\theta L_\theta(\theta), & (184)\\ &= (\nabla_\theta \eta)^{-1} (\nabla_\theta \eta) \nabla_\eta L_\eta(\eta), & (185)\\ &= \nabla_\eta L_\eta(\eta). & (186) \end{aligned}$$

□

It follows that the natural gradient descent on a loss function $L_\theta(\theta)$ amounts to an ordinary gradient descent on the *dually parameterized* loss function $L_\eta(\eta) := L_\theta(\theta(\eta))$. In short, ${}^{\text{NG}}\nabla_\theta L_\theta = \nabla_\eta L_\eta$.

4.1.4 An application of the natural gradient: Natural Evolution Strategies (NESs)

A nice family of applications of the natural gradient are the Natural Evolution Strategies (NESs) for black-box minimization [19]: Let $f(x)$ for $x \in \mathbb{X} \subset \mathbb{R}^d$ be a real-valued function to minimize. Berny [18] proposed to *relax* the optimization problem $\min_{x\in\mathbb{X}} f(x)$ by considering a parametric search distribution p_λ, and minimize instead:

$$\min_{\lambda\in\Lambda} E_{p_\lambda}[f(x)], \tag{187}$$

where $\lambda \in \Lambda \subset \mathbb{R}^D$ denotes the parameter space of the search distributions. Let $J(\lambda) = E_{p_\lambda}[f(x)]$. Minimizing $J(\lambda)$ instead of $f(x)$ is particularly useful when $\mathbb{X}$ is a discrete space: Indeed, the *combinatorial*

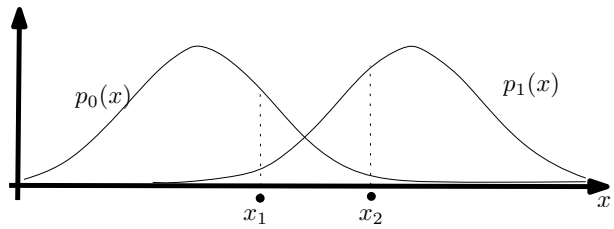

Figure 11: Statistical Bayesian hypothesis testing: The best Maximum A Posteriori (MAP) rule chooses to classify an observation from the class that yields the maximum likelihood.

optimization [18] $\min_{x\in\mathbb{X}} f(x)$ is replaced by a continuous optimization $\min_{\lambda\in\Lambda} J(\lambda)$ when Λ is a continuous parameter, and the ordinary or natural GD methods can be used. The gradient $\nabla J(\lambda)$ is called the *search gradient*, and it can be approximated stochastically using the log-likelihood trick [135] as

$$\tilde{\nabla}\ J(\lambda) := \frac{1}{n}\sum_{i=1}^{n} f(x_i)\nabla \log p_\lambda(x_i) \approx \nabla J(\lambda), \tag{188}$$

where $x_1,\ldots,x_n \sim p_\lambda$. Similarly, the Fisher information matrix (FIM) may be approximated by the following empirical FIM:

$$\tilde{I}(\lambda) = \frac{1}{n}\sum_{i=1}^{n} \nabla_\lambda l_\lambda(x_i)(\nabla_\lambda l_\lambda(x_i))^\top \approx I(\lambda), \tag{189}$$

where $l_\lambda(x) := \log p_\lambda(x)$ denote the log-likelihood function. Notice that the approximated FIM may potentially be degenerated and may not respect the structure of the true FIM. For example, we have $\nabla_\mu l(x;\mu,\sigma^2) = \frac{x-\mu}{\sigma^2}$ and $\nabla_{\sigma^2} = \frac{(x-\mu)^2}{2\sigma^4} - \frac{1}{2\sigma^2}$. The non-diagonal of the approximate FIM $\tilde{I}(\lambda)$ are close to but usually non-zero although the expected FIM is diagonal $I(\mu,\sigma^2) = \mathrm{diag}\left(\frac{1}{\sigma^2}, \frac{1}{2\sigma^4}\right)$. Thus we may estimate the FIM until the non-diagonal elements have absolute values less than a prescribed $\epsilon > 0$. For multivariate normals, we have $\nabla_\mu l(x;\mu,\Sigma) = \Sigma^{-1}(x-\mu)$ and $\nabla_\Sigma l(x;\mu,\Sigma) = \frac{1}{2}(\nabla_\mu l(x;\mu,\Sigma)\nabla_\mu l(x;\mu,\Sigma)^\top - \Sigma^{-1})$.

4.2 Some illustrating applications of dually flat manifolds

In this part, we describe how to use the dually flat structures for handling an exponential family $\mathcal{E}$ (in a hypothesis testing problem detailed in §4.3) and the mixture family $\mathcal{M}$ (clustering statistical mixtures §4.4). Note that for a general divergence, neither $(\mathcal{E}, D)$ nor $(\mathcal{M}, D)$ is dually flat. However, when $D = \mathrm{KL}$, the Kullback-Leibler divergence, we get dually flat spaces that are computationally attractive since the primal/dual geodesics are straight lines in the corresponding global affine coordinate system.

4.3 Hypothesis testing in the dually flat exponential family manifold $(\mathcal{E}, \mathrm{KL}^*)$

Given two probability distributions $P_0 \sim p_0(x)$ and $P_1 \sim p_1(x)$, we ask to classify a set of iid. observations $X_{1:n} = \{x_1,\ldots,x_n\}$ as either sampled from P_0 or from P_1? This is a statistical decision problem [73]. For example, P_0 can represent the signal distribution and P_1 the noise distribution. Figure 11 displays the probability distributions and the unavoidable error that is made by any statistical decision rule (on observations x_1 and x_2).

Assume that both distributions $P_0 \sim P_{\theta_0}$ and $P_1 \sim P_{\theta_1}$ belong to the same *exponential family* $\mathcal{E} = \{P_\theta : \theta \in \Theta\}$, and consider the exponential family manifold with the dually flat structure $(\mathcal{E}, {}_{\mathcal{E}}g, {}_{\mathcal{E}}\nabla^e, {}_{\mathcal{E}}\nabla^m)$. That is, the manifold equipped with the Fisher information metric tensor field and the expected exponential connection and conjugate expected mixture connection. More generally, the expected α-geometry of an